

EE320 - Electronics II

Cascode Current Source Design and Analysis in LTspice

0.18 μm CMOS Technology

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Circuit: NMOS cascode current source
Software: LTspice

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1. Objective

The objective of this project is to design and analyze an NMOS cascode current source using LTspice and a provided $0.18\ \mu\text{m}$ CMOS model. The circuit uses a diode-connected NMOS transistor to generate the reference gate voltage, an NMOS current mirror to copy the reference current, and a cascode NMOS transistor to improve the output resistance.

The required specifications are:

Table 1: Project specifications.

Parameter	Value
Supply voltage, V_{DD}	1.8 V
Reference current, I_{REF}	0.5 mA
Technology	$0.18\ \mu\text{m}$ CMOS
Transistor type	NMOS
Transistor dimensions	$W = 5\ \mu\text{m}$, $L = 0.18\ \mu\text{m}$
Target output current	$I_{out} = 0.5\ \text{mA}$

2. Circuit Description

The circuit is shown conceptually in Fig. 1. Transistor M_1 is diode-connected because its drain and gate are connected together. The reference current I_{REF} forces a gate-source voltage across M_1 . This voltage is then applied to the gate of M_2 , making M_2 mirror the reference current.

Transistor M_3 is the cascode device. Its gate is biased by V_b . The cascode transistor raises and stabilizes the internal node voltage V_x between M_2 and M_3 . This reduces the effect of output voltage changes on the drain-source voltage of M_2 , which improves the output resistance.

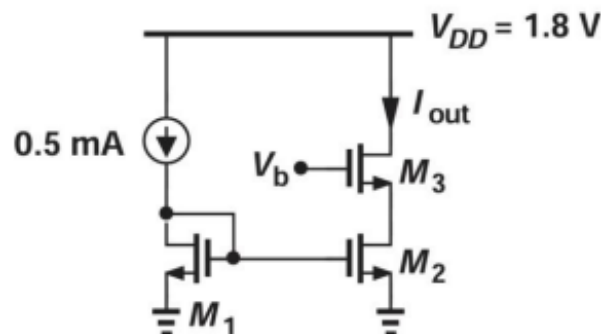


Figure 1: Reference cascode current source circuit from the project statement.

3. LTspice Implementation

The provided NMOS symbol and the 0.18 μm CMOS model file were used. The model name used for the NMOS devices was `nfet`. Each transistor was assigned:

$$W = 5\text{ }\mu\text{m}, \quad L = 0.18\text{ }\mu\text{m} \quad (1)$$

The main LTspice directives used for the operating point simulations were:

```
.include tsmc018.lib
.param VB=1.8
.op
```

For the output resistance simulation, the drain of M_3 was connected to a separate voltage source called `VOUT`. The output voltage was swept using:

```
.param VB=1.8
.dc VOUT 0 1.8 0.01
```

The implemented LTspice schematic used for output resistance extraction is shown in Fig. 2.

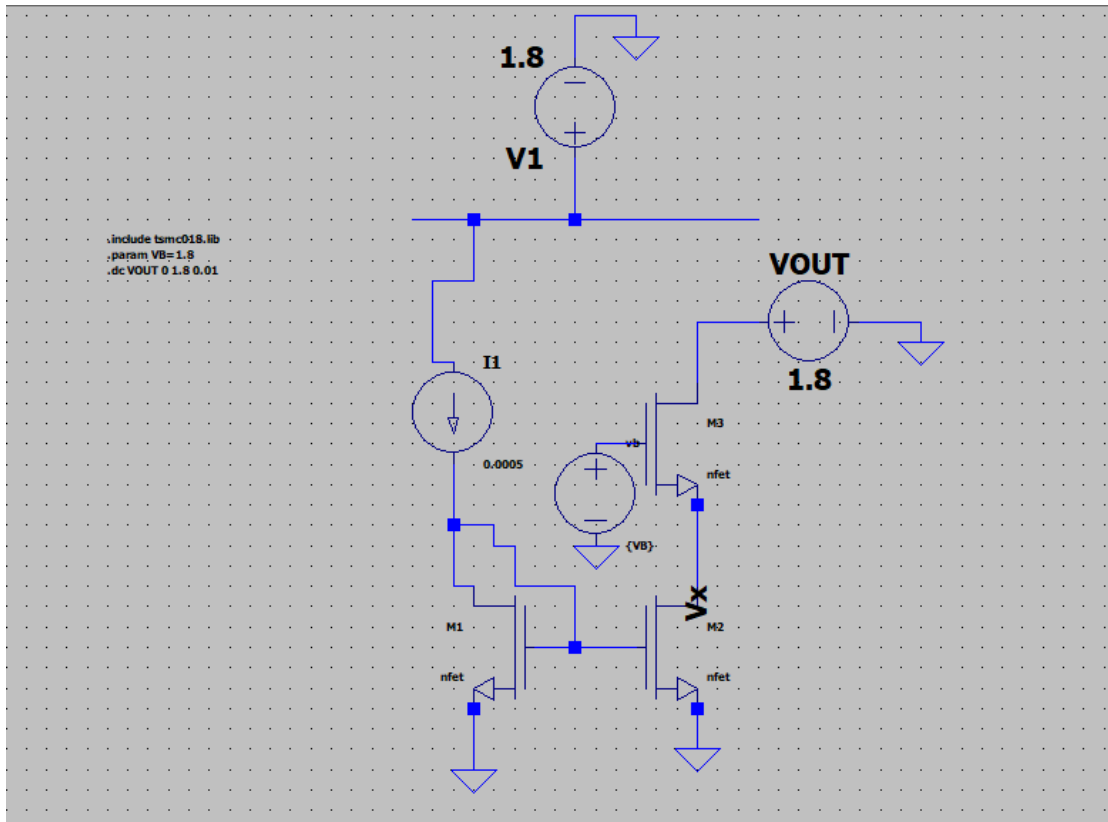


Figure 2: LTspice schematic used for the output resistance sweep. The output voltage source `VOUT` is connected to the drain of M_3 .

4. Hand Analysis for Bias Voltage

4.1. Current Mirror Operation

Since M_1 and M_2 have the same dimensions and share the same gate-source voltage, the ideal current mirror relation is:

$$I_{D2} \approx I_{D1} \quad (2)$$

Since the reference current is:

$$I_{D1} = I_{REF} = 0.5 \text{ mA} \quad (3)$$

then ideally:

$$I_{D2} \approx 0.5 \text{ mA} \quad (4)$$

This ideal relation assumes that both M_1 and M_2 are in saturation and that channel-length modulation is neglected.

4.2. MOSFET Saturation Requirement

For an NMOS transistor to operate in saturation:

$$V_{DS} \geq V_{GS} - V_{TH} \quad (5)$$

The overdrive voltage is:

$$V_{OV} = V_{GS} - V_{TH} \quad (6)$$

For M_2 , the drain-source voltage is the internal node voltage:

$$V_{DS2} = V_x \quad (7)$$

Therefore, M_2 needs:

$$V_x \geq V_{OV2} \quad (8)$$

The cascode transistor M_3 controls V_x . Increasing V_b increases the gate voltage of M_3 , which raises V_x and helps M_2 remain in saturation.

4.3. Approximate Bias Relation

For M_3 :

$$V_{GS3} = V_b - V_x \quad (9)$$

To conduct approximately the same current as M_2 , M_3 needs a suitable gate-source voltage. Using the square-law approximation:

$$I_D = \frac{1}{2}k'_n \frac{W}{L} V_{OV}^2 \quad (10)$$

Solving for the overdrive voltage:

$$V_{OV} = \sqrt{\frac{2I_D}{k'_n(W/L)}} \quad (11)$$

Then:

$$V_{GS} = V_{TH} + V_{OV} \quad (12)$$

A simple hand estimate for the cascode bias is:

$$V_b \approx V_x + V_{GS3} \quad (13)$$

If V_x is selected approximately equal to the required saturation voltage of M_2 :

$$V_x \approx V_{OV2} \quad (14)$$

and since M_2 and M_3 carry approximately the same current:

$$V_{OV2} \approx V_{OV3} \quad (15)$$

then:

$$V_b \approx V_{TH} + 2V_{OV} \quad (16)$$

This is a theoretical estimate. The actual value was selected using LTspice because the 0.18 μm model includes short-channel and non-ideal effects.

5. Selection of Bias Voltage V_b

Several values of V_b were tested. The selected nominal value was:

$$V_{b0} = 1.8 \text{ V} \quad (17)$$

At this value, the operating point result gave:

$$I_{out} = I_D(M3) = 0.500 \text{ 448 mA} \quad (18)$$

This is very close to the required current of 0.5 mA. The nominal operating point screenshot is shown in Fig. 3.

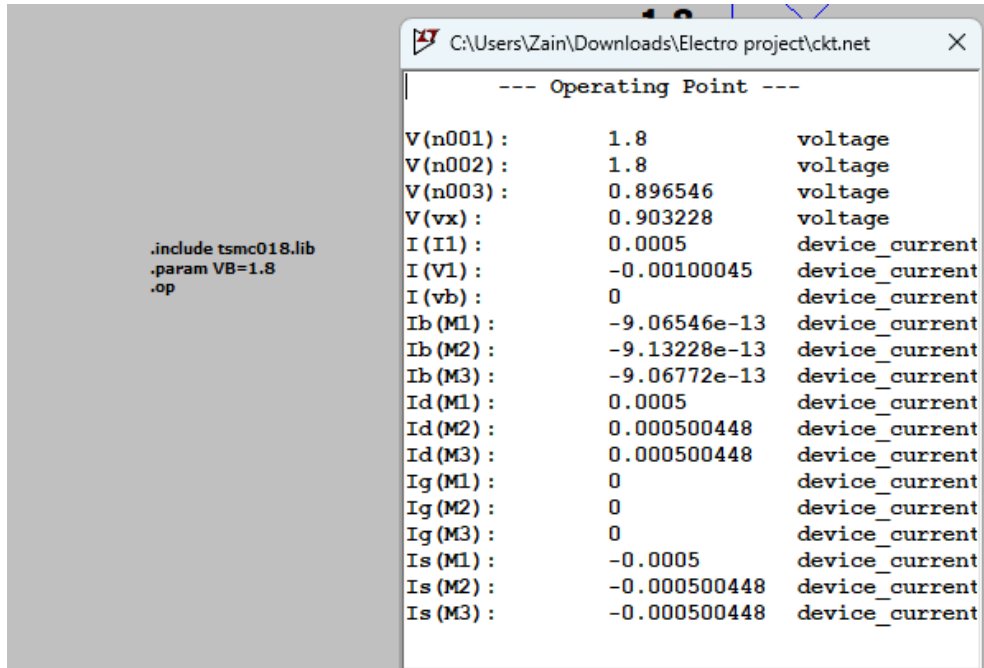


Figure 3: Operating point result at $V_b = 1.8$ V. The output current is approximately 0.500 448 mA.

Important node voltages at the nominal bias are:

Table 2: Nominal operating point at $V_b = 1.8$ V.

Quantity	Value
V_{DD}	1.8 V
V_b	1.8 V
V_g	0.895 46 V
V_x	0.903 28 V
$I_D(M1)$	0.500 000 mA
$I_D(M2)$	0.500 448 mA
$I_D(M3)$	0.500 448 mA

6. Effect of Bias Voltage Variation

The project requires testing the effect of changing V_b by ± 100 mV around the nominal value. Since:

$$V_{b0} = 1.8 \text{ V} \quad (19)$$

The tested values were:

$$V_b = 1.7 \text{ V}, \quad 1.8 \text{ V}, \quad 1.9 \text{ V} \quad (20)$$

The simulation results are summarized in Table 3.

Table 3: Effect of V_b variation on output current.

Case	V_b	V_x	$I_{out} = I_D(M3)$
$V_{b0} - 100$ mV	1.7 V	0.809 75 V	0.494 057 mA
Nominal	1.8 V	0.903 28 V	0.500 448 mA
$V_{b0} + 100$ mV	1.9 V	0.996 60 V	0.506 600 mA

The corresponding operating point screenshots are shown in Fig. 4 and Fig. 5.

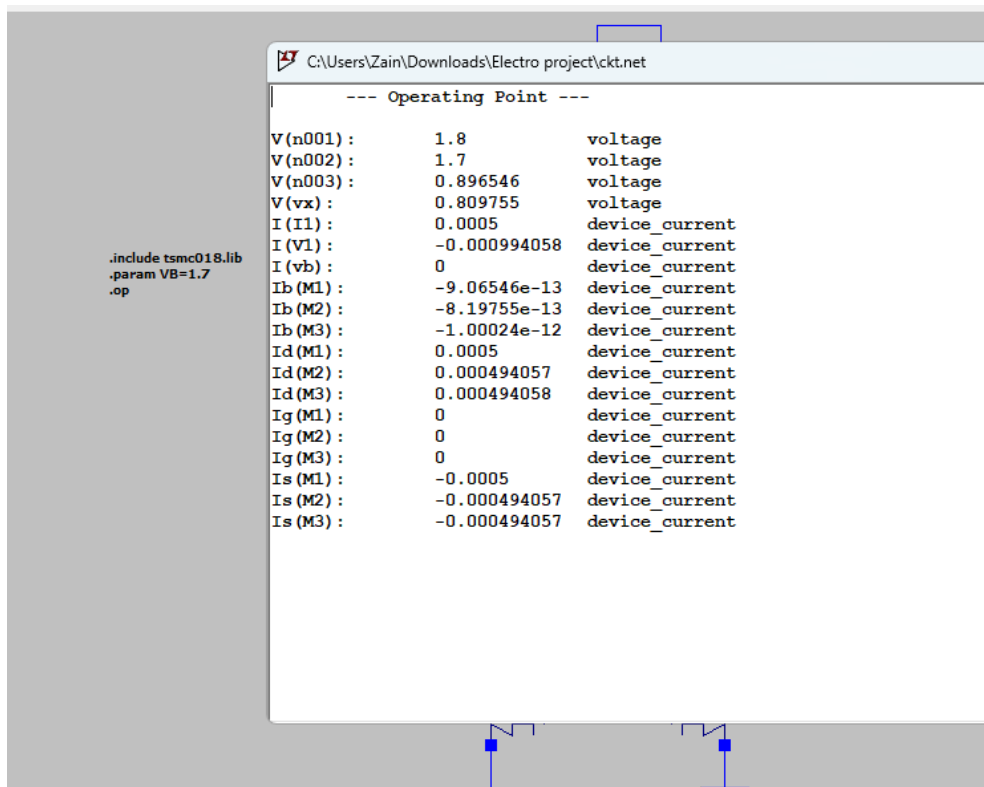


Figure 4: Operating point result at $V_b = 1.7$ V. The output current is 0.494 057 mA.

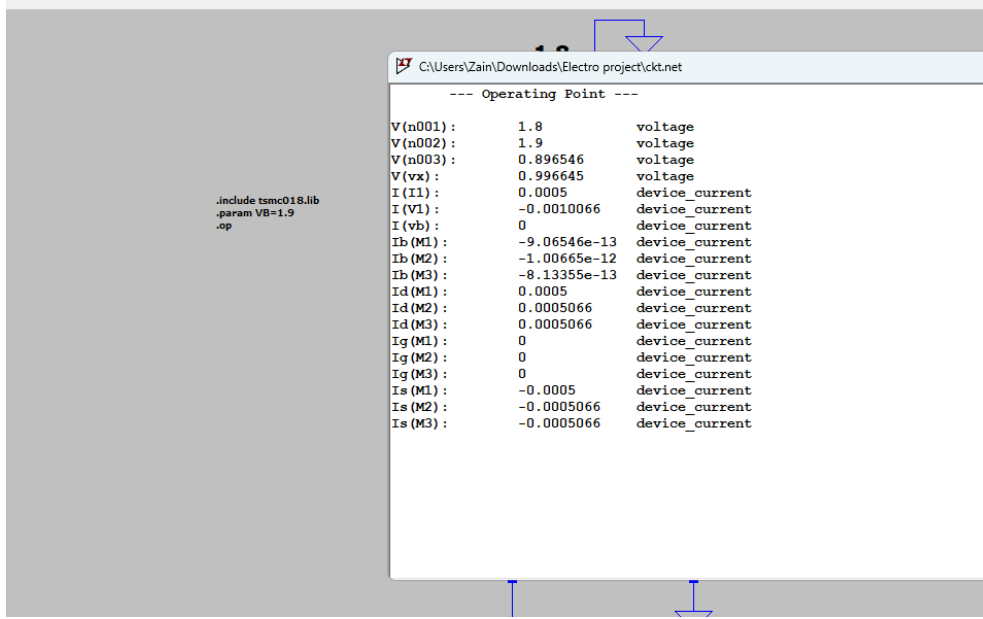


Figure 5: Operating point result at $V_b = 1.9$ V. The output current is 0.506 600 mA.

6.1. Discussion of V_b Variation

Increasing V_b increases the gate voltage of M_3 . This increases the source voltage of M_3 , which is the internal node V_x . Therefore, V_x rises as V_b increases.

When V_x increases, the drain-source voltage of M_2 also increases. This helps M_2 operate more strongly in saturation and makes the mirrored current closer to the reference current. The output current therefore increases slightly as V_b increases.

The current does not remain perfectly constant because real MOSFETs are not ideal. Channel-length modulation and short-channel effects cause the drain current to change slightly with drain-source voltage. The cascode transistor reduces this effect, but it does not remove it completely.

7. Output Resistance Analysis

7.1. Small-Signal Derivation

The output resistance is defined as:

$$R_{out} = \frac{\Delta V_{out}}{\Delta I_{out}} \quad (21)$$

For a simple current mirror without a cascode transistor, the output resistance is approximately:

$$R_{out} \approx r_{o2} \quad (22)$$

where r_{o2} is the small-signal output resistance of M_2 .

For the cascode current source, the output resistance is increased by the cascode transistor M_3 . Neglecting body effect, the small-signal output resistance is approximately:

$$R_{out} \approx r_{o2} + r_{o3} + g_{m3}r_{o2}r_{o3} \quad (23)$$

where:

$$r_{o2} = \frac{1}{g_{ds2}}, \quad r_{o3} = \frac{1}{g_{ds3}} \quad (24)$$

Usually, the product term is dominant:

$$g_{m3}r_{o2}r_{o3} \gg r_{o2} + r_{o3} \quad (25)$$

Therefore:

$$R_{out} \approx g_{m3}r_{o2}r_{o3} \quad (26)$$

This equation shows the main benefit of the cascode transistor. The output resistance is not only r_{o2} anymore. It is multiplied by a factor related to $g_{m3}r_{o3}$, so the output current becomes much less sensitive to changes in V_{out} .

7.2. LTspice Extraction of Output Resistance

To extract R_{out} from LTspice, V_b was fixed at 1.8 V. Then the output voltage source VOUT was swept from 0 V to 1.8 V. The simulated output current curve is shown in Fig. 6.

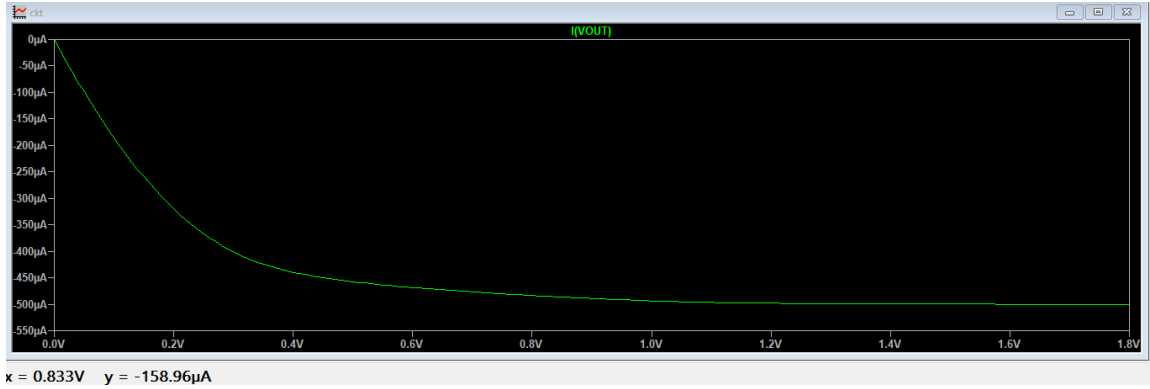


Figure 6: Output current $I(VOUT)$ versus output voltage. At low V_{out} , the current source does not have enough voltage headroom. At higher V_{out} , the current becomes almost flat.

From the flat region of the curve, two points were selected:

$$V_{out1} = 1.1996 \text{ V}, \quad I_{out1} = 498.22 \mu\text{A} \quad (27)$$

$$V_{out2} = 1.7990 \text{ V}, \quad I_{out2} = 501.54 \mu\text{A} \quad (28)$$

The output resistance is:

$$R_{out} = \frac{V_{out2} - V_{out1}}{I_{out2} - I_{out1}} \quad (29)$$

Substituting the simulated values:

$$R_{out} = \frac{1.7990 - 1.1996}{(501.54 - 498.22) \times 10^{-6}} \quad (30)$$

$$R_{out} = \frac{0.5994}{3.32 \times 10^{-6}} \quad (31)$$

$$R_{out} \approx 180\,542\,\Omega \quad (32)$$

Therefore:

$$\boxed{R_{out,sim} \approx 180.5\,\text{k}\Omega} \quad (33)$$

8. Comparison Between Theory and Simulation

The theoretical expression predicts a high output resistance because the cascode transistor increases the resistance seen looking into the drain of M_3 . The simulated result confirms this behavior. In the output current plot, the current becomes nearly constant after the circuit enters its proper operating region.

The output resistance extracted from LTspice was:

$$R_{out,sim} \approx 180.5\,\text{k}\Omega \quad (34)$$

The analytical expression was kept symbolic because the exact numerical values of g_m and g_{ds} depend on the detailed $0.18\,\mu\text{m}$ transistor model. The simulation includes short-channel effects, channel-length modulation, and model parameters that are not included in the simple square-law hand calculation. Therefore, exact matching between hand analysis and LTspice is not expected.

The important agreement is qualitative: the cascode device greatly improves output resistance and makes the output current much less sensitive to V_{out} .

9. Handwritten Solution Summary

This section is written in a compact form so it can be copied by hand.

1. Current Mirror

Since M_1 and M_2 are identical:

$$\frac{W_1}{L_1} = \frac{W_2}{L_2}$$

and they share the same gate-source voltage:

$$V_{GS1} = V_{GS2}$$

Therefore, ideally:

$$I_{D2} = I_{D1} = I_{REF} = 0.5 \text{ mA}$$

2. Saturation Conditions

For NMOS saturation:

$$V_{DS} \geq V_{GS} - V_{TH} = V_{OV}$$

For M_2 :

$$V_{DS2} = V_x$$

So:

$$V_x \geq V_{OV2}$$

For M_3 :

$$V_{GS3} = V_b - V_x$$

So:

$$V_b = V_x + V_{GS3}$$

Using the square-law model:

$$I_D = \frac{1}{2} k'_n \frac{W}{L} V_{OV}^2$$

$$V_{OV} = \sqrt{\frac{2I_D}{k'_n(W/L)}}$$

$$V_{GS} = V_{TH} + V_{OV}$$

Approximate cascode bias:

$$V_b \approx V_{TH} + 2V_{OV}$$

From LTspice tuning:

$$V_{b0} = 1.8 \text{ V}, \quad I_{out} = 0.500448 \text{ mA}$$

3. Output Resistance

Small-signal output resistance:

$$R_{out} \approx r_{o2} + r_{o3} + g_{m3}r_{o2}r_{o3}$$

Since the product term is dominant:

$$R_{out} \approx g_{m3}r_{o2}r_{o3}$$

From simulation:

$$R_{out} = \frac{\Delta V_{out}}{\Delta I_{out}}$$

$$R_{out} = \frac{1.7990 - 1.1996}{(501.54 - 498.22) \times 10^{-6}}$$

$$R_{out} \approx 180.5 \text{ k}\Omega$$

10. How the Circuit Can Be Improved

The circuit works correctly and reaches the required output current. However, it can be improved in several ways:

1. **Use a better bias generator for V_b .** In this simulation, V_b was manually selected. In a real integrated circuit, V_b should be generated using a bias circuit so that it tracks process, voltage, and temperature changes.
2. **Increase channel length.** Using a longer channel length for M_2 and M_3 would reduce channel-length modulation and increase output resistance.
3. **Use a wide-swing cascode.** The circuit has limited voltage headroom because V_{DD} is only 1.8 V. A wide-swing cascode structure can improve output swing while keeping high output resistance.
4. **Use layout matching techniques.** In practical IC layout, M_1 and M_2 should be placed close to each other with common-centroid or interdigitated layout to improve current matching.
5. **Add startup and bias stability considerations.** A complete real design should make sure the bias network starts correctly and remains stable under all operating conditions.

11. Conclusion

An NMOS cascode current source was implemented and analyzed in LTspice using the provided 0.18 μm CMOS model. The selected bias voltage was:

$$V_{b0} = 1.8 \text{ V}$$

At this bias, the output current was:

$$I_{out} = 0.500448 \text{ mA}$$

which satisfies the required 0.5 mA target. Changing V_b by $\pm 100 \text{ mV}$ changed the output current by only about $6 \mu\text{A}$, showing that the cascode structure stabilizes the current. The simulated output resistance was extracted from the I_{out} versus V_{out} curve and was found to be:

$$R_{out,sim} \approx 180.5 \text{ k}\Omega$$

The results confirm that the cascode transistor improves the current source by increasing output resistance and reducing the dependence of output current on output voltage.